MODEL LAB · 2026-10-05

Every prior, same slate

The held-out scores that decided which prior is served, then how each one would have ranked today's hitters.

Served (full season)
served prior

11
priors compared

33,641
held-out plate appearances

3.05%
held-out HR rate

0.5999
best AUC · KNN comps

Held-out scores

Fitted on plate appearances before 2026-08-26, scored on the 33,641 that came after (actual HR rate 3.05%). No outcome in the test window was visible during fitting. **Read AUC first** — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors. Metrics are per plate appearance; the probabilities lower down are per game.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5861	1.53×	0.02951	0.13549	0.00512
KNN comps Prior is the pooled rate of the nearest comparables.	0.5999	1.52×	0.02947	0.13464	0.00463
SVR Support-vector regression on the same feature matrix.	0.5948	1.53×	0.02951	0.13536	0.00592
Random forest Forest regression; a learned similarity metric.	0.5943	1.52×	0.02950	0.13499	0.00508
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.5999	1.52×	0.02947	0.13464	0.00463
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5999	1.52×	0.02947	0.13464	0.00463
Linear regression Ordinary least squares on the same features.	0.5945	1.54×	0.02949	0.13506	0.00414
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5951	1.62×	0.02951	0.13510	0.00626
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5944	1.55×	0.02950	0.13501	0.00571
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5951	1.52×	0.02951	0.13508	0.00592

BLEND WEIGHTS (FOREST VARIANT)

KNN **1.00** SVR **0.00** RF **0.00**

SHRINKAGE K PER PRIOR

league **248** knn **248** svr **248** rf **248** core **248** ensemble **248** linear **248** logistic **248** nn **248** xgb **248**

Estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-sample subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (3,663 PA, actual HR rate 2.46%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5600	1.00×	0.02398	0.11553	0.00855
KNN comps Prior is the pooled rate of the nearest comparables.	0.5712	1.33×	0.02405	0.11639	0.01603
SVR Support-vector regression on the same feature matrix.	0.5748	1.22×	0.02421	0.12004	0.01766
Random forest Forest regression; a learned similarity metric.	0.5818	1.33×	0.02415	0.11709	0.01569
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.5712	1.33×	0.02405	0.11639	0.01603
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5712	1.33×	0.02405	0.11639	0.01603
Linear regression Ordinary least squares on the same features.	0.5728	1.22×	0.02414	0.11860	0.01591
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5768	1.45×	0.02414	0.11770	0.01556
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5730	1.33×	0.02410	0.11730	0.01548
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5829	1.33×	0.02421	0.11778	0.01841

Slate agreement

PRIOR	SHARED	ITS TOP SIX
Served (full season)	6/6	Ben Rice 19.7%, Junior Caminero 16.7%, Luis García Jr. 16.6%, Munetaka Murakami 16.5%, Spencer Jones 15.9%, George Lombard Jr. 14.8%
Flat league	6/6	Ben Rice 18.2%, Luis García Jr. 16.1%, Junior Caminero 16.0%, George Lombard Jr. 15.1%, Munetaka Murakami 14.9%, Spencer Jones 14.6%
KNN comps	6/6	Ben Rice 19.4%, Luis García Jr. 16.9%, Junior Caminero 16.5%, Munetaka Murakami 16.4%, Spencer Jones 15.6%, George Lombard Jr. 14.8%
SVR	6/6	Ben Rice 20.0%, Junior Caminero 17.7%, Luis García Jr. 17.6%, Munetaka Murakami 17.1%, George Lombard Jr. 14.8%, Spencer Jones 14.3%
Random forest	6/6	Ben Rice 19.8%, Luis García Jr. 17.2%, Junior Caminero 16.9%, Munetaka Murakami 16.6%, George Lombard Jr. 14.8%, Spencer Jones 14.8%
Core (KNN+SVR)	6/6	Ben Rice 19.4%, Luis García Jr. 16.9%, Junior Caminero 16.5%, Munetaka Murakami 16.4%, Spencer Jones 15.6%, George Lombard Jr. 14.8%
Forest blend	6/6	Ben Rice 19.4%, Luis García Jr. 16.9%, Junior Caminero 16.5%, Munetaka Murakami 16.4%, Spencer Jones 15.6%, George Lombard Jr. 14.8%
Linear regression	6/6	Ben Rice 19.5%, Luis García Jr. 17.0%, Junior Caminero 16.7%, Munetaka Murakami 16.3%, Spencer Jones 14.9%, George Lombard Jr. 14.9%
Logistic regression	6/6	Ben Rice 19.7%, Luis García Jr. 17.4%, Junior Caminero 17.0%, Munetaka Murakami 16.5%, Spencer Jones 15.0%, George Lombard Jr. 14.2%
Neural net (torch)	6/6	Ben Rice 19.7%, Luis García Jr. 17.4%, Junior Caminero 17.1%, Munetaka Murakami 16.6%, Spencer Jones 15.3%, George Lombard Jr. 14.5%
XGBoost	6/6	Ben Rice 19.8%, Luis García Jr. 17.4%, Junior Caminero 17.3%, Munetaka Murakami 17.0%, Spencer Jones 15.3%, George Lombard Jr. 14.8%

Hitter by hitter

The top 40 bats by the served probability.

HITTER	SERVED (FULL SEASON)	FLAT LEAGUE	KNN COMPS	SVR	RANDOM FOREST	CORE (KNN+SVR)	FOREST BLEND	LINEAR REGRESSION	LOGISTIC REGRESSION	NEURAL NET (T
1 **Ben Rice** NYY slate	**19.7%**	18.2%	19.4%	20.0%	19.8%	19.4%	19.4%	19.5%	19.7%	
2 **Junior Caminero** TB slate	**16.7%**	16.0%	16.5%	17.7%	16.9%	16.5%	16.5%	16.7%	17.0%	
3 **Luis García Jr.** NYY	**16.6%**	16.1%	16.9%	17.6%	17.2%	16.9%	16.9%	17.0%	17.4%	
4 **Munetaka Murakami** NYY slate	**16.5%**	14.9%	16.4%	17.1%	16.6%	16.4%	16.4%	16.3%	16.5%	
5 **Spencer Jones** NYY	**15.9%**	14.6%	15.6%	14.3%	14.8%	15.6%	15.6%	14.9%	15.0%	
6 **George Lombard Jr.** NYY	**14.8%**	15.1%	14.8%	14.8%	14.8%	14.8%	14.8%	14.9%	14.2%	
7 **Jonathan Aranda** TB	**13.6%**	12.8%	13.7%	13.4%	13.4%	13.7%	13.7%	13.5%	13.3%	
8 **Miguel Vargas** CWS	**13.3%**	12.0%	13.3%	13.3%	13.1%	13.3%	13.3%	13.3%	13.5%	
9 **Cody Bellinger** NYY	**12.7%**	13.1%	12.9%	12.4%	12.8%	12.9%	12.9%	12.8%	12.8%	
10 **Kyle Teel** CWS	**12.3%**	12.0%	12.5%	12.1%	12.4%	12.5%	12.5%	12.4%	11.5%	
11 **Yandy Díaz** TB	**11.8%**	12.1%	11.8%	12.2%	12.0%	11.8%	11.8%	12.1%	11.9%	
12 **Jazz Chisholm Jr.** NYY	**11.7%**	11.2%	11.9%	11.4%	11.5%	11.9%	11.9%	11.5%	11.5%	
13 **Jo Adell** LAA slate	**11.6%**	11.0%	11.6%	11.9%	11.6%	11.6%	11.6%	11.5%	11.9%	
14 **Andrew Benintendi** CWS	**11.5%**	10.3%	11.2%	10.8%	11.0%	11.2%	11.2%	11.3%	11.5%	
15 **Austin Wells** NYY	**11.2%**	11.0%	11.3%	11.0%	11.0%	11.3%	11.3%	11.2%	11.0%	
16 **Victor Mesa Jr.** TB	**10.8%**	10.7%	11.1%	11.2%	11.1%	11.1%	11.1%	11.0%	11.0%	
17 **Colson Montgomery** CWS	**10.0%**	9.0%	10.3%	9.8%	10.1%	10.3%	10.3%	10.3%	10.8%	
18 **Ryan McMahon** NYY	**9.7%**	9.0%	9.5%	9.3%	9.5%	9.5%	9.5%	9.5%	9.3%	

	HITTER		SERVED (FULL SEASON)	FLAT LEAGUE	KNN COMPS	SVR	RANDOM FOREST	CORE (KNN+SVR)	FOREST BLEND	LINEAR REGRESSION	LOGISTIC REGRESSION	NEURAL NET (T
19	**Chase DeLauter**	CLE	**8.9%**	9.0%	8.8%	8.7%	8.6%	8.8%	8.8%	8.7%	8.6%	
20	**Nathaniel Lowe**	CLE	**8.9%**	8.5%	8.8%	8.7%	9.1%	8.8%	8.8%	8.7%	8.7%	
21	**Tristan Peters**	CWS	**8.7%**	9.2%	8.8%	9.0%	8.9%	8.8%	8.8%	8.9%	9.0%	
22	**Angel Martínez**	CLE	**8.5%**	8.7%	8.3%	8.7%	8.5%	8.3%	8.3%	8.4%	8.6%	
23	**Jonny DeLuca**	TB	**8.0%**	8.6%	8.0%	8.0%	7.5%	8.0%	8.0%	7.8%	8.1%	
24	**Sam Antonacci**	CWS	**7.7%**	9.3%	8.0%	7.2%	8.1%	8.0%	8.0%	7.8%	7.7%	
25	**Patrick Bailey**	CLE	**7.7%**	7.6%	8.2%	7.9%	7.7%	8.2%	8.2%	7.9%	7.5%	
26	**Travis Bazzana**	CLE	**7.3%**	7.4%	7.1%	7.0%	7.3%	7.1%	7.1%	7.1%	6.9%	
27	**Braden Montgomery**	CWS	**7.2%**	8.1%	6.8%	6.7%	6.7%	6.8%	6.8%	6.7%	6.6%	
28	**Liam Hicks**	TB	**7.1%**	8.0%	6.9%	7.7%	7.4%	6.9%	6.9%	7.4%	7.3%	
29	**José Ramírez**	CLE	**7.0%**	7.3%	7.2%	6.7%	7.1%	7.2%	7.2%	7.3%	7.5%	
30	**Chase Meidroth**	CWS	**6.7%**	7.5%	6.7%	7.0%	7.0%	6.7%	6.7%	6.9%	6.7%	
31	**Richie Palacios**	TB	**6.1%**	6.7%	6.0%	5.8%	5.7%	6.0%	6.0%	5.9%	5.8%	
32	**Brayan Rocchio**	CLE	**5.3%**	5.7%	5.2%	5.3%	5.3%	5.2%	5.2%	5.1%	5.0%	
33	**Taylor Walls**	TB	**3.9%**	5.5%	4.2%	3.5%	3.5%	4.2%	4.2%	4.1%	4.1%	
34	**Steven Kwan**	CLE	**3.4%**	6.2%	3.7%	2.7%	2.8%	3.7%	3.7%	2.0%	3.4%	
35	**Chandler Simpson**	TB	**2.2%**	4.9%	2.8%	1.4%	1.4%	2.8%	2.8%	1.2%	2.6%	

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context. Shading grows with the probability.

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